Meridian Junior College
JC2 Preliminary Examinations

Name:	
Class:	Index No:

BIOLOGY 9747/03

Paper 3 12 Sep 2008

1 hour 30 minutes

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and register number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of examination,
1. Write your name, index number and civics group on all sheets of writing papers.
2. Fasten all your writing papers together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
Total	

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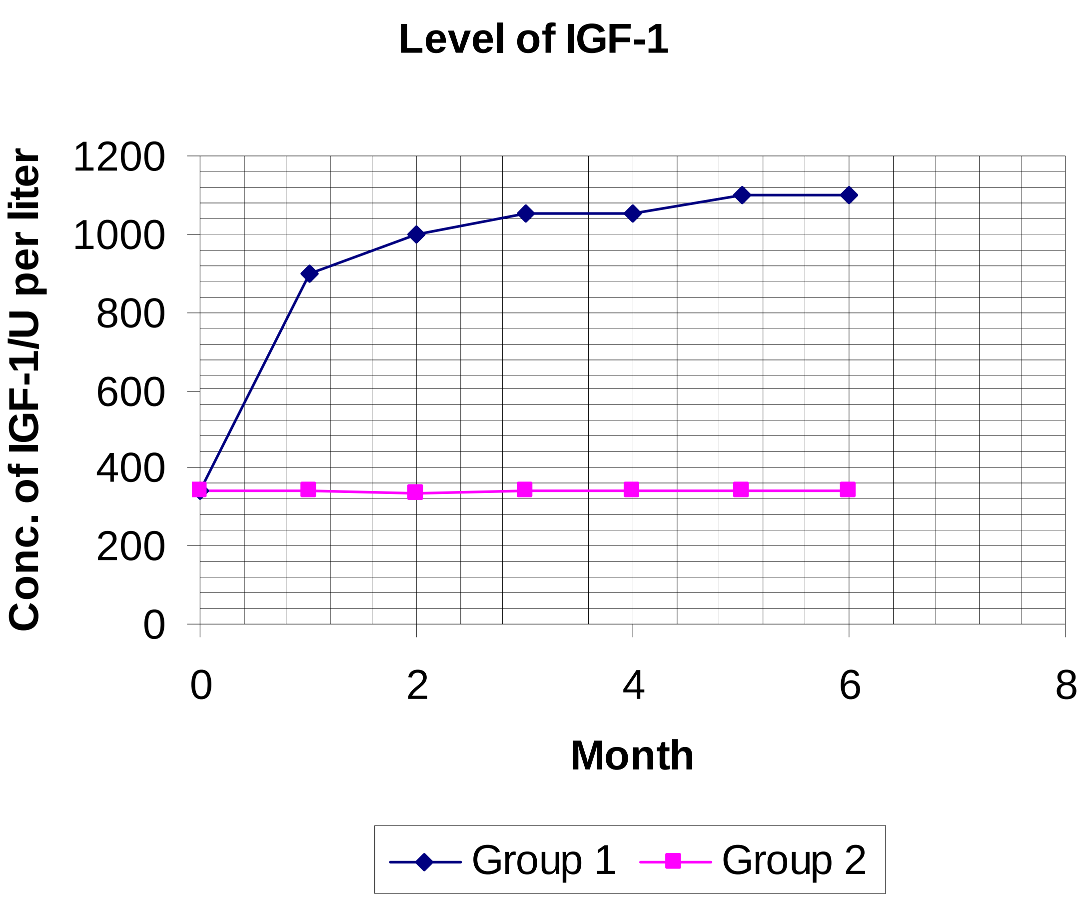
1. In a study on the effect of Human Growth Hormone (HGH) in aging men over 60 years old, the level of growth hormone-insulin-like growth factor 1 (IGF-1) in the plasma of 21 healthy men from 61 to 81 years old were studied during a six-month base-line period and a six-month treatment period that followed. The men all had plasma IGF-1 concentrations of less than 350 U per liter.

   Growth hormone secretion can be measured indirectly, however, by measuring the plasma concentration of IGF-I, which is produced and released by the liver in response to growth hormone.

   During the treatment period, 12 men (group 1) received approximately 0.03 mg of biosynthetic HGH per kilogram of body weight subcutaneously three times a week, and 9 men (group 2) received no treatment. Plasma IGF-1 levels were measured monthly. At the end of each period, the lean body mass and mass of adipose tissue were measured. The results were shown in Graph 1.1 and Table 1.2 below.



Graph 1.1

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Month		0	1	2	3	4	5	6
Group 1	Accumulative % change in lean body mass	0	+2.6	+3.0	+4.7	+6.2	+7.2	+8.8
	Accumulative % change in adipose tissue mass	0	-1.4	-2.9	-4.5	-8.0	-11.7	-14.4
Group 2	Accumulative % change in lean body mass	0	0	0	0	0	0	0
	Accumulative % change in adipose tissue mass	0	0	0	0	0	0	0

Table 1.2

(a) Comment on the effect of human growth hormone on the aging men. [3]

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The results suggested that the atrophy of the lean body mass and the enlargement of the mass of adipose tissue that are characteristic of the elderly resulted at least in part from diminished secretion of growth hormone. If so, the age-related changes in body composition should be correctable in part by the administration of human growth hormone, now readily available as a biosynthetic product.

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(b) Suggest how recombinant HGH plasmids can be produced via genetic engineering. [4]

(c) The recombinant HGH plasmids are inserted into bacteria host cells via transformation. Describe the transformation steps. [2]

The transformed bacteria were cultured on an agar plate. The plasmid pBR322 was used for the production of recombinant plasmids. There are two antibiotic resistance genes in the plasmid, coding for ampicillin resistance and tetracycline resistance. The restriction enzyme used for the cleaving of the plasmid was AluI and would cleave at the restriction site as shown in Fig 1.3.

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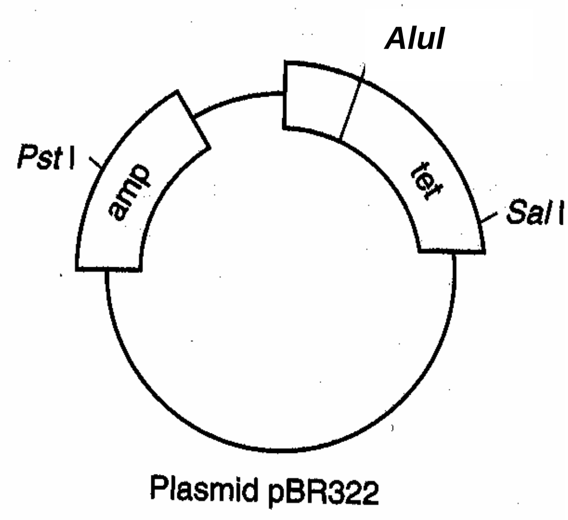

Fig 1.3

(d) Suggest the selection medium/media to use in the agar plate(s) for selection of the bacteria with recombinant plasmids. [1]

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(e) Outline the steps that will be taken to culture the bacteria on the plate(s) for the selection of the bacteria with recombinant plasmids. [3]

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(f) Suggest what can be done to ascertain the gene in the transformed bacteria with recombinant plasmid is HGH gene? [2]

[Total: 15 marks]

2. Cord blood is the blood that circulates through the umbilical cord from the foetus to the placenta. The umbilical cord that remains attached to the placenta has been found to be rich in blood stem cells. Blood stem cells are the young or immature cells that can transform into other forms of essential blood cell types, such as red blood cells, white blood cells and platelets.

Traditionally, the main source of blood stem cells still comes from bone marrow. However, extraction of cells from bone marrow requires a surgical procedure that can be painful and may carry (although minimal) risk to the donor. Cord blood cells, on the other hand, are much easier to collect through a simple procedure. Blood is drawn from the umbilical vein, which takes place after birth and after the umbilical cord is separated from the newborn baby. Therefore, this simple cord blood collection procedure involves no risk to the baby or mother.

HLA, or histo-compatibility antigens, which represent immune genotypes, are present on the surface of all cells. The foetus and mother will be of different HLA type. Yet, the foetal cells are immune tolerated by its mother. HLA typing for histo-compatibility testing is used for determining the compatibility of the stem cells (whether bone marrow or cord blood stem cells) of the donor for the recipient. Because of its immunological tolerance, cord blood stem cells can allow for a lower degree of HLA matching as compared to bone marrow (which requires an exact match). Therefore there is a greater chance for a patient to find a match within a cord blood bank compared to a bone marrow registry. There is a decreased risk of graft-vs-host disease for transplantation using cord blood, as compared to using adult bone marrow.

Umbilical cord blood was first used for transplantation in 1988, in France, for a patient with Fanconi´s anemia. Since then, cord blood has been increasingly used as a substitute for bone marrow in thousands of transplants worldwide. Several previously life-threatening diseases such as SCID (Severe Combined Immune Deficiency), leukemias, lymphomas, thalassemia major and severe aplastic anemia can now be cured using stem cell transplantation.

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(a) Describe the unique features of all stem cells. [3]

(b) Explain the advantages of using cord blood stem cells in place of bone marrow. [2]

Cord blood stem cells are regarded as multipotent stem cells that have the potential to differentiate into a limited number of cell types. There exists other groups of stem cells of greater developmental potential.

(c) Describe with an example for each, the two groups of stem cells with higher developmental potential than cord blood stem cells. [2]

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Besides using stem cell transplant to treat genetic disorders such as SCID, gene therapy could be another treatment option. Severe Combined Immunodeficiency (SCID) is a disease of young children in which patients have neither cell-mediated immune responses nor the ability to make antibodies. There are two types of SCID: Adenosine Deaminase (ADA) deficient SCID and X-linked SCID. There are also two types of delivery system used in SCID gene therapy: non-viral gene delivery system and viral gene delivery system.

(d) Suggest a suitable viral vector used in SCID gene therapy. [1]

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(e) Outline the use of viral gene delivery system in gene therapy. [3]

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X-linked SCID is caused by a mutated X-linked gene encoding a subunit called the gamma c (γc) of the receptor for several interleukins including interleukin-7 (IL-7). IL-7 is essential in converting blood stem cells into precursors of T cells. A non-viral gene delivery system is available to specifically target the mutated gene for the treatment of X-linked SCID.

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(f) Describe the non-viral gene delivery system used in the gene therapy of X-linked SCID which specifically targets the mutated gene. [4]

[Total: 15 marks]

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3. Plant growth regulators (PGR) such as auxin, cytokinin and ethylene are naturally produced in plants and are important in determination of the developmental pathway of plant cells. Cheaper synthetic analogues are frequently used in plant tissue culture instead of the natural PGR. Table 3.1 summarizes the general characteristics and roles of some natural and synthetic PGR.

Type of PGR	**Examples**	**Role**
Auxins	IAA(natural) - unstable to heat & light 2,4-D(synthetic) – stable to heat & light	Promote cell division and cell growth Root initiation (when auxin:cytokinin is high)
Cytokinins	2iP(natural) – unstable to heat & light Kinetin(synthetic) - stable to heat & light	Promotes cell division Shoot formation (when auxin:cytokinin is low)
Abscisic acid (ABA)	-	Inhibits cell division Maturation of somatic embryos Abscission of plant leaves Seed dormancy Induces stomatal closure to reduce water loss by transpiration
Ethylene	-	Abscission of leaves and flowers Seed and bud dormancy by growth inhibition Fruit ripening

Table 3.1

Prunus lannesiana is an early-flowering cherry (only in spring) in Japan Izu peninsula. It is commonly known as Sakura or Japanese Cherry. Researchers are keen to propagate *P. lannesiana* from sterilized explants by micropropagation due to the advantages that the technique offers. Hence, a study was made to analyse the concentration of PGR present in the *P. lannesiana* in the four seasons and the

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results were summarized in Table 3.2. Table 3.3 illustrates the physiological development of *P. lannesiana* in the various seasons.

Season / Conc. of PGR /arbitrary unit	Spring (Mar-May)	Summer (June – Aug)	Autumn (Sep-Nov)	Winter (Dec – Feb)
IAA (auxin)	15	20	25	30
2iP (cytokinin)	10	15	10	5
ABA	5	5	12	15
Ethylene	12 – 20	18	16	14

Table 3.2

Season	**Spring**	**Summer**	**Autumn**	**Winter**
Shoot development	Very Active	Active	Minimal	Nil
Root development	Minimal	Moderate	Active	Very Active
Leaves development	Very Active	Active	Senescence and abscission	Nil
Flowers development	Short full bloom in early spring, followed by senescence	Nil	Nil	Buds develop but dormant in late winter
Fruit and seed development	Fruit and seed development in late spring	Seed maturation	Seed dormant	Seed dormant

Table 3.3

(a) State two limitations of micropropagation. [2]

(b) Suggest why synthetic PGR (e.g. 2,4-D and Kinetin) are used instead of natural PGR (e.g. IAA and 2iP) in plant tissue culture. [1]

(c) (i) Using the information provided in Table 3.2 and Table 3.3, explain the effect of the change in the auxin:cytokinin ratio from Spring to Winter and vice versa. [4]

(c) (ii) Using the information provided in Table 3.2 and Table 3.3, explain the effect of a change in concentration of ABA from Spring to Winter. [4]

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A gene encoding ethylene inhibitor protein has been isolated from a related species and researchers would like to clone this gene into *P. lannesiana* in an effort to extend the blooming period. Fig 3.4 illustrates a Ti plasmid which is commonly used to transfer genes into plants using the soil bacterium *Agrobacterium tumefaciens* as a vector.

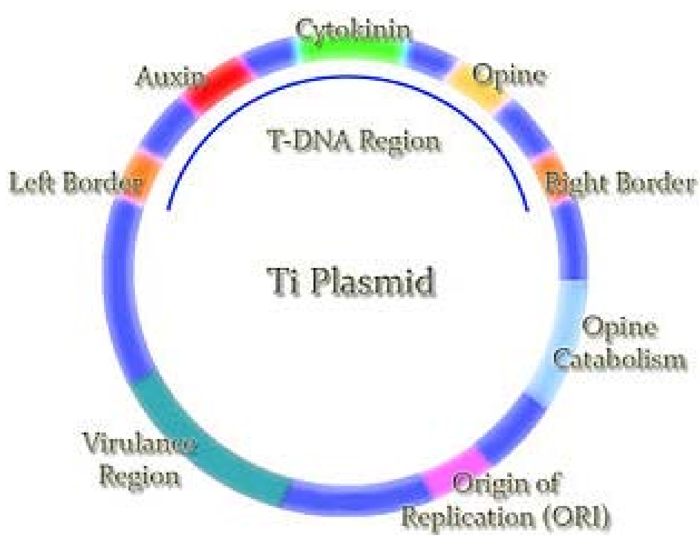

Fig 3.4

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(d) Describe how the gene of interest can be inserted into Ti plasmid to form a recombinant Ti plasmid. [4]

[Total: 15 marks]

Write your answers to this question on the writing paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a)**, **(b)** etc, as indicated by the question.

4. (a) Describe how nucleic acid hybridization can be used in RFLP analysis on isolated DNA sample. [6]

(b) Explain the application of RFLP analysis in disease detection and DNA fingerprinting. [8]

(c) Explain how gel electrophoresis is used to analyse nucleic acids. [6]